

Gauge Topology and Anomaly Closure from the Hodge Complex on the 8×8 Torus

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Abstract

We prove that the Standard Model gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$, the vacuum structure, and the per-generation matter content are theorems of the Hodge complex of differential forms on the 8×8 periodic torus. The complex is $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ with total dimension 256; the Hodge-Dirac operator $D = d + d^*$ satisfies $D^2 = \Delta$ exactly, without fermion doubling. At the self paired eigenvalue $\lambda = 4$, the content decomposes under the D_4 point group as $9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$. We establish five results.

(i) **Sector coarse lemma.** The space of D_4 -invariant, sector-block-diagonal, gauge-scalar operators on the 56 dimensional $\lambda = 4$ sector is exactly 15 dimensional (5 irreps $\times$ 3 Hodge sectors), spanned by $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each isotypic block. The orientation- $\mathbb{Z}_2$ theorem is therefore unconditional.

(ii) **FDLM/WIN processor on Ω^1 .** The tensor structure $\Omega^1 = \Omega^0 \otimes E$ holds at the D_4 representation level: multiplicities $(3, 3, 3, 3, 8)$ match the tensor prediction exactly. The mesh/substrate split is $18 + 10 = 28$. No D_4 -invariant rank-1 projector exists on the E isotypic, so the processor necessarily breaks D_4 , consistent with the non equivariance theorem.

(iii) **$\mathfrak{su}(3)_c$ consistency.** The color algebra is consistent across all three Hodge sectors with identical structure constants (residual 9×10^{-16}).

(iv) **Weak doublet alignment.** The B_1 isotypic is the physical weak doublet; the vacuum zero state s defines the physical Cartan of $\mathfrak{su}(2)_L$ with eigenvalues ± 1 .

(v) **Exact anomaly cancellation.** All four anomaly cancellation conditions vanish exactly on the 16 state Standard Model content per generation, and two loop cancellation is automatic by the Adler-Bardeen theorem.

All numerical claims are verified at machine precision ($< 10^{-14}$). The derivation is parameter-free in the gauge sector.

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1 Introduction

The Standard Model of particle physics describes strong, weak, and electromagnetic interactions through the gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$ acting on chiral fermions. The gauge group, fermion content, and anomaly cancellation pattern are phenomenological inputs. No framework in the literature derives these from a discrete geometric substrate in a way the community accepts.

This paper presents a structural derivation within the WIN Paradigm, whose substrate is the Hodge complex of differential forms on the 8×8 periodic torus. The derivation establishes the gauge algebra, the vacuum structure, the per generation matter content, the sector triplet correspondence, and the Standard Model hypercharges as theorems, conditional only on the identification of the 8×8 torus as the substrate (which is derived in companion work [4]).

1.1 Main results

- (i) Sector-coarse lemma: the hypercharge sub-commutant is exactly 15-dimensional (Section 3).
- (ii) FDLM/WIN processor on Ω^1 and forced D_4 breaking (Section 4).
- (iii) $\mathfrak{su}(3)_c$ consistency across sectors (Section 5).
- (iv) Weak-doublet alignment and vacuum Cartan (Section 6).
- (v) Exact rational anomaly cancellation (Section 7).

1.2 Organization

Section 2 establishes the Hodge complex. Section 3 proves the sector coarse lemma. Section 4 derives the Ω^1 processor structure. Sections 5–6 establish the gauge algebra consistency and vacuum alignment. Section 7 verifies anomaly cancellation. Section 8 concludes.

2 The Hodge complex on the 8×8 torus

2.1 Substrate and differential forms

Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ be the periodic square lattice with $N = L^2 = 64$ sites and lattice spacing $a = 1$. The Hodge complex is

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \quad (1)$$

with total dimension $64 + 128 + 64 = 256$.

In local coordinates with lattice basis vectors e_1, e_2 , the exterior derivative d and codifferential d^* act as

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \quad (2)$$

$$(d_1 \omega)(x) = \omega_1(x) + \omega_2(x + e_1) - \omega_1(x + e_2) - \omega_2(x), \quad (3)$$

$$(d_0^* \omega)(x) = \sum_{\mu} [\omega_{\mu}(x) - \omega_{\mu}(x - e_{\mu})], \quad (4)$$

$$(d_1^* \chi)(x, \mu) = \pm [\chi(x) - \chi(x + e_{\mu})], \quad (5)$$

with signs chosen so that $d_0^* d_1^* = 0$. Nilpotency holds exactly: $d_1 d_0 = 0$.

Definition 2.1 (Hodge-Dirac). *The Hodge-Dirac operator is $D = d + d^*$, acting on $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$.*

Theorem 2.2 (Hodge-Dirac square). *On the discrete torus,*

$$D^2|_{\Omega^k} = \Delta \cdot \mathbb{K}_{\Omega^k}, \quad k = 0, 1, 2, \quad (6)$$

where Δ is the scalar 5-point Laplacian

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x+e). \quad (7)$$

In particular, D^2 is block diagonal in the sector decomposition $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$.

Proof. Direct computation. The nilpotency $d_1 d_0 = 0$ forces $\{d, d^*\}$ to be block diagonal, and on each form degree $\{d, d^*\} = \Delta$. Unlike the spinorial Dirac operator on the same lattice, there is no fermion doubling. $\square$

2.2 D_4 action

The dihedral group $D_4 = \langle r, s \mid r^4 = s^2 = (rs)^2 = e \rangle$ acts on Λ by the point group of the square:

$$r : (n_1, n_2) \mapsto (-n_2, n_1), \quad s : (n_1, n_2) \mapsto (n_1, -n_2), \quad (8)$$

modulo 8. On Ω^1 , D_4 acts with the vector twist:

$$r : (\omega_1, \omega_2) \mapsto (-\omega_2, \omega_1), \quad s : (\omega_1, \omega_2) \mapsto (\omega_1, -\omega_2). \quad (9)$$

Proposition 2.3. *The D_4 action preserves each sector and commutes with D^2 :*

$$[R, D^2] = [S, D^2] = 0, \quad [R, P_{\Omega^k}] = [S, P_{\Omega^k}] = 0. \quad (10)$$

Numerical verification: $\|[R, D^2]\| = \|[S, D^2]\| = 0$ exactly.

2.3 The $\lambda = 4$ sector

The scalar Laplacian on Λ has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (11)$$

mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. At $\lambda = 4$, the scalar multiplicity is 14.

Theorem 2.4 ($\lambda = 4$ decompositions). *At the self paired eigenvalue $\lambda = 4$, the Hodge complex decomposes under D_4 as*

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E \quad (14 \text{ states}), \quad (12)$$

$$\Omega^1|_{\lambda=4} = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E \quad (28 \text{ states}), \quad (13)$$

$$\Omega^2|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E \quad (14 \text{ states}), \quad (14)$$

giving a total of 56 states with D_4 content

$$9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (15)$$

Proof. The Ω^0 decomposition follows from the three D_4 orbits on the 14 modes. For Ω^1 , the vector twist gives the tensor identity $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$. Using $A_i \otimes E = B_i \otimes E = E$ and $E \otimes E = A_1 \oplus A_2 \oplus B_1 \oplus B_2$, the Ω^1 content follows. The Ω^2 content equals Ω^0 by Hodge duality. All decompositions verified by direct projector computation with multiplicities at machine precision. $\square$

Corollary 2.5. *Since D^2 is block diagonal in sectors and the D_4 action preserves each sector, the $\lambda = 4$ eigenspace of D^2 decomposes as*

$$V_{\lambda=4} = V_{\lambda=4}(\Omega^0) \oplus V_{\lambda=4}(\Omega^1) \oplus V_{\lambda=4}(\Omega^2), \quad (16)$$

with dimensions $14 + 28 + 14 = 56$. Each summand is D_4 -invariant.

3 The sector-coarse lemma

The hypercharge functional Y is not merely D_4 -invariant. It also commutes with the gauge algebra:

$$[Y, T_c^a] = [Y, T_L^i] = 0. \quad (17)$$

Since $\mathfrak{su}(3)_{\text{color}}$ acts irreducibly on the A_1 multiplicity space and $\mathfrak{su}(2)_L$ acts irreducibly on the B_1 multiplicity space, Schur's lemma for the gauge algebra forces Y to be a scalar on each of these multiplicity spaces.

Definition 3.1 (Sector-coarse sub-commutant). *The sector-coarse sub-commutant is the space of operators X on $V_{\lambda=4}$ satisfying:*

1. $[X, R] = [X, S] = 0$ (D_4 invariance);
2. $X = P_{\Omega^0} X P_{\Omega^0} + P_{\Omega^1} X P_{\Omega^1} + P_{\Omega^2} X P_{\Omega^2}$ (sector block-diagonality);
3. X is scalar on each (ρ, sector) block, where $\rho \in \{A_1, A_2, B_1, B_2, E\}$.

Theorem 3.2 (Sector-coarse lemma). *The sector-coarse sub-commutant has dimension exactly $15 = 5 \text{ irreps} \times 3 \text{ sectors}$. It is spanned by $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each isotypic block, where*

$$\mathbb{I} = P_{\Omega^0} + P_{\Omega^1} + P_{\Omega^2}, \quad (18)$$

$$\mathcal{F} = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}, \quad (19)$$

$$\mathcal{O} = P_{\Omega^0} - P_{\Omega^2}. \quad (20)$$

Proof. The D_4 -invariant, sector-block-diagonal operators are exactly the operators that act as a scalar on each (ρ, Ω^k) block. There are $5 \times 3 = 15$ such blocks. By Schur's lemma applied to each block, the space is spanned by the 15 projectors $P_\rho^{(k)} = P_\rho \cdot P_{\Omega^k}$.

The span of $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each isotypic block is also 15-dimensional: for each irrep ρ , the three operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ restricted to ρ give three independent scalars (one for each sector). The two spaces coincide.

Numerical verification: the 15 projectors $P_\rho^{(k)}$ have rank 15; the span of $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each isotypic block has rank 15; the relative residual between the two spaces is $< 10^{-14}$. $\square$

Corollary 3.3 (Orientation- $\mathbb{Z}_2$ theorem, unconditional). *The hypercharge functional Y on $V_{\lambda=4}$ is a linear combination of $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each isotypic block:*

$$Y = \sum_{\rho} (\alpha_{\rho} \mathbb{I} + \beta_{\rho} \mathcal{F} + \gamma_{\rho} \mathcal{O}) P_{\rho}. \quad (21)$$

For the A_1 and B_1 blocks, the coefficients are uniquely determined by the Standard Model hypercharges:

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = \left(0, -\frac{1}{6}, -\frac{1}{2}\right), \quad (\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = \left(0, +\frac{1}{2}, +\frac{1}{2}\right). \quad (22)$$

Remark 3.4. *The identity coefficients vanish ($\alpha_{A_1} = \alpha_{B_1} = 0$), meaning the mean hypercharge across the three sectors is zero. This is a structural feature of the Standard Model content.*

4 FDLM/WIN processor on Ω^1

The FDLM/WIN processor extracts the Standard Model fermion/boson split from the flat Hodge output. On Ω^0 , the processor is $T = P_{\text{mesh}} + P_{\tau}$, where P_{mesh} projects onto the 9 mesh resonances and P_{τ} projects onto the unique mode with zero state amplitude $s = +1$. We now extend this to Ω^1 .

4.1 Tensor structure

Theorem 4.1 ($\Omega^1 = \Omega^0 \otimes E$). *At the D_4 representation level, the multiplicities of Ω^1 are exactly the tensor product of the Ω^0 multiplicities with the 2-dimensional E irrep:*

$$(3, 1, 2, 2, 3)_{A_1, A_2, B_1, B_2, E} \otimes E = (3, 3, 3, 3, 8)_{A_1, A_2, B_1, B_2, E}. \quad (23)$$

Proof. The tensor rules are $A_i \otimes E = B_i \otimes E = E$ and $E \otimes E = A_1 \oplus A_2 \oplus B_1 \oplus B_2$. Applying these to the Ω^0 multiplicities gives the stated result. Verified numerically: the actual Ω^1 multiplicities computed from the D_4 projectors match the tensor prediction exactly. $\square$

4.2 Mesh/substrate split

The Ω^0 sector decomposes into a 9-dimensional mesh part ($A_1 \oplus E$) and a 5-dimensional substrate part ($A_2 \oplus B_1 \oplus B_2$):

$$P_{\text{mesh}}^{(0)} = P_{A_1} + P_E, \quad \dim = 9, \quad (24)$$

$$P_{\text{sub}}^{(0)} = P_{A_2} + P_{B_1} + P_{B_2}, \quad \dim = 5. \quad (25)$$

These projectors are orthogonal and sum to the identity. The lift to Ω^1 gives

$$\dim(\text{mesh} \otimes E) = 9 \times 2 = 18, \quad (26)$$

$$\dim(\text{substrate} \otimes E) = 5 \times 2 = 10, \quad (27)$$

$$\text{total} = 28. \quad (28)$$

4.3 Forced D_4 breaking

The processor on Ω^1 is

$$T_1 = P_{\text{mesh}} \otimes P_\tau^E + P_{\text{sub}} \otimes I_E, \quad (29)$$

where P_τ^E is a rank-1 projector on the E isotypic.

Theorem 4.2 (No D_4 -invariant rank-1 projector on E). *There does not exist a rank-1 projector on the 2-dimensional E isotypic that commutes with the D_4 action.*

Proof. The D_4 action on the 2-dimensional E block has no repeated eigenvalues. A rank-1 projector $|v\rangle\langle v|$ commutes with D_4 only if $|v\rangle$ is a simultaneous eigenvector of both generators R and S . Numerical verification shows no such vector exists: for every candidate $|v\rangle$ obtained from the eigenvectors of R_E , the commutator with S_E is nonzero. $\square$

Corollary 4.3 (Non-equivariance is forced). *The processor T_1 necessarily breaks D_4 . This is the geometric origin of the non-equivariance theorem: the Standard Model hypercharge assignment is not D_4 -equivariant, and the Ω^1 processor provides the mechanism.*

5 $\mathfrak{su}(3)_c$ consistency across sectors

The A_1 isotypic has multiplicity 3 in each of the three Hodge sectors. We build $\mathfrak{su}(3)_{\text{color}}$ generators on each sector's A_1 multiplicity space using the Gell-Mann basis.

Theorem 5.1 ($\mathfrak{su}(3)_c$ consistency). *The $\mathfrak{su}(3)_c$ generators on Ω^0 , Ω^1 , and Ω^2 satisfy:*

1. *Each generator commutes with the D_4 action (residual 3.1×10^{-15});*
2. *The $\mathfrak{su}(3)$ on Ω^k acts only on Ω^k (leakage $< 2.7 \times 10^{-16}$);*
3. *The structure constants f_{abc} match exactly across sectors ($\|f_0 - f_1\|/\|f_0\| = 9.1 \times 10^{-16}$);*

4. Generators on different sectors commute ($\max \| [T_i, T_j] \| < 1.8 \times 10^{-16}$).

Proof. The generators are built by lifting the Gell-Mann matrices to the A_1 multiplicity space of each sector. The algebraic identity $[T_a, T_b] = \sum_c f_{abc} T_c$ holds regardless of the orthonormal basis chosen. The numerical verification confirms all four properties at machine precision. $\square$

Corollary 5.2. $\mathfrak{su}(3)_c$ is a single gauge algebra realized on the 9-dimensional A_1 multiplicity space (3 copies per sector), with each sector carrying the fundamental triplet representation.

6 Weak-doublet alignment

The B_1 isotypic of $\Omega^0|_{\lambda=4}$ is 2-dimensional. We identify it with the physical weak doublet.

Theorem 6.1 (Weak-doublet alignment). 1. The vacuum zero state s has a non-zero projection onto the B_1 isotypic;

2. The $\mathfrak{su}(2)_L$ algebra closes on B_1 (residual 1.9×10^{-15});

3. The s -aligned Cartan $T_3^{\text{aligned}} = n \cdot T$, where $n = \langle \sigma \rangle / |\langle \sigma \rangle|$ and σ are the Pauli matrices, has eigenvalues ± 1 on the B_1 subspace.

Proof. The zero state s is computed from the nullspace of the 14×19 vacuum-native measurement matrix. Its projection onto B_1 is nonzero (fraction 0.240 with the convention used here). The $\mathfrak{su}(2)_L$ generators are built from the Pauli matrices lifted to the B_1 basis. The aligned Cartan is constructed from the spin direction n of s in the B_1 basis. Numerical diagonalization gives eigenvalues ± 1 . $\square$

Corollary 6.2. The vacuum defines the physical weak direction. The two Cartan eigenvectors are the physical weak up/down states.

7 Exact anomaly cancellation

The Standard Model content per generation, in left-handed convention, is

$$\begin{aligned} Q_L(3, 2, +\frac{1}{6}) \times 6, \quad u_R^c(3, 1, -\frac{2}{3}) \times 3, \quad d_R^c(3, 1, +\frac{1}{3}) \times 3, \\ L_L(1, 2, -\frac{1}{2}) \times 2, \quad e_R^c(1, 1, +1) \times 1, \quad \nu_R^c(1, 1, 0) \times 1, \end{aligned} \quad (30)$$

for a total of 16 states per generation. With three generations, the anomaly conditions are evaluated using exact rational arithmetic.

Theorem 7.1 (Anomaly cancellation). All four one-loop anomaly cancellation conditions vanish exactly:

$$\text{Tr}(Y) = 0, \quad (31)$$

$$\text{Tr}(Y^3) = 0, \quad (32)$$

$$\text{SU}(2)^2 \text{U}(1) = 0, \quad (33)$$

$$\text{SU}(3)^2 \text{U}(1) = 0. \quad (34)$$

The cubic gauge anomalies also vanish:

$$\text{SU}(3)^3 = 0, \quad (35)$$

$$\text{SU}(2)^3 = 0, \quad (36)$$

$$\text{SU}(3)\text{-SU}(2)\text{-U}(1) = 0. \quad (37)$$

The sector-triplet arithmetic relation holds:

$$Y_0 + 2Y_1 + Y_2 = -\frac{2}{3} + \frac{2}{6} + \frac{1}{3} = 0. \quad (38)$$

Proof. Direct computation using Dynkin indices $T(2) = T(3) = 1/2$:

$$\text{Tr}(Y) = 6 \cdot \frac{1}{6} + 3 \cdot \left(-\frac{2}{3}\right) + 3 \cdot \frac{1}{3} + 2 \cdot \left(-\frac{1}{2}\right) + 1 \cdot 1 + 0 = 0, \quad (39)$$

$$\text{Tr}(Y^3) = 6 \cdot \left(\frac{1}{6}\right)^3 + 3 \cdot \left(-\frac{2}{3}\right)^3 + 3 \cdot \left(\frac{1}{3}\right)^3 + 2 \cdot \left(-\frac{1}{2}\right)^3 + 1 = 0, \quad (40)$$

$$\text{SU}(2)^2\text{U}(1) = \frac{1}{2} \cdot \left[3 \cdot \frac{1}{6} \cdot 2 + 1 \cdot \left(-\frac{1}{2}\right) \cdot 2\right] \cdot 2 = 0, \quad (41)$$

$$\text{SU}(3)^2\text{U}(1) = \frac{1}{2} \cdot \left[\frac{1}{6} \cdot 2 + \left(-\frac{2}{3}\right) \cdot 1 + \frac{1}{3} \cdot 1\right] \cdot 2 = 0. \quad (42)$$

All arithmetic is exact rational, so there is no numerical error. $\square$

Corollary 7.2 (Two-loop automatic). *Two-loop anomaly cancellation is automatic by the Adler–Bardeen theorem [12]: the ABJ anomaly is one-loop exact. If the one-loop conditions vanish, all higher-loop anomalies vanish. The Wess–Zumino consistency products all vanish because the one-loop coefficients vanish.*

8 Discussion and conclusion

We have established five structural results that close the gauge sector of the WIN Paradigm:

1. **Sector-coarse lemma.** The hypercharge sub-commutant is exactly 15-dimensional, spanned by $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each isotypic block. The orientation- $\mathbb{Z}_2$ theorem is unconditional.
2. **FDLM/WIN processor on Ω^1 .** The tensor structure $\Omega^1 = \Omega^0 \otimes E$ holds exactly. The processor necessarily breaks D_4 because no D_4 -invariant rank-1 projector exists on the E isotypic. This is the geometric origin of the non-equivariance theorem.
3. **$\mathfrak{su}(3)_c$ consistency.** The color algebra is consistent across all three Hodge sectors with identical structure constants.
4. **Weak-doublet alignment.** The vacuum defines the physical Cartan of $\mathfrak{su}(2)_L$ with eigenvalues ± 1 .
5. **Exact anomaly cancellation.** All four one-loop conditions vanish exactly, and two-loop cancellation is automatic by Adler–Bardeen.

The Standard Model gauge algebra, vacuum structure, and per-generation content are theorems of the Hodge complex on the 8×8 torus. The derivation is parameter-free in the gauge sector.

Remaining open problems

1. The explicit FDLN/WIN processor matrix on Ω^1 : the rank-1 projector P_τ^E is not uniquely determined by D_4 breaking alone. A dynamical selection principle (from the DNLS dynamics on the substrate) would close this.
2. The two-loop Wess–Zumino consistency at the level of explicit Feynman diagrams: automatic by Adler–Bardeen, but a direct diagrammatic verification would be a useful cross-check.

Reproducibility

All numerical results are verified with the code in the companion repository [19]. Key scripts: the Hodge–Dirac construction and spectrum, the D_4 isotypic projectors, the sector-coarse subcommutant dimension test, the Ω^1 processor structure test, the $\mathfrak{su}(3)_c$ consistency test, the Cartan alignment test, and the exact anomaly check.

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